

Find the amplitude, period, phase shift, vertical shift, domain, and range for each of the following functions.

1. $y = -5 \cos[2(x - \pi)] - 10$

2. $y = 4 \sin \frac{1}{3} \left(x - \frac{\pi}{4} \right)$

Graph one period of the following functions. Then, identify the domain, range, amplitude, period, phase shift, and vertical shift.

3. $y = \cos(3x + \pi)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

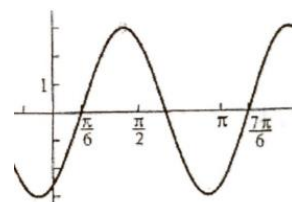
4. $y = 3 \cos(2x + 5\pi)$

- Amplitude
- Period
- Phase shift
- Vertical shift
- Domain
- Range
- Frequency

5. Fill out the characteristics. Then, write a cosine equation to represent the functions graphed.

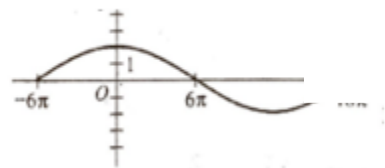
a. Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



b. Amplitude _____ Period _____

Phase shift _____ Vertical shift _____



6. Use the given characteristics to write an equation.

a. Cosine curve with amplitude = $\frac{1}{2}$, period = $\frac{\pi}{2}$, phase shift = none, and vertical shift = up 7

b. Cosine curve with amplitude = $\frac{2}{3}$, period = π , phase shift = $-\frac{\pi}{4}$, and vertical shift = none.